

Checkpoint 1 Submission: Bitcoin OHLCV Dataset

(A) Identification of Candidate Datasets

1) Bitcoin OHLCV Price/Volume Dataset

- **Dataset Name & Source:** [Kaggle: Bitcoin Historical Data \(SwapTr\)](#)
- **Course Topic Alignment:**
 - Week 5: Large-Scale ML - modeling/predicting price and volume
 - Week 12: Stream Mining - analyze rolling windows of high-frequency price/volume data
 - Week 9: Anomaly Detection - detect unusual price/volume spikes
- **Beyond-Course Techniques (Not Covered in Weeks 1–15):**
 - LSTM/GRU or Transformer-based time-series forecasting
 - Change-point detection for structural breaks in price
 - Technical indicator-based predictive modeling (like RSI, Bollinger Bands)
- **Dataset Size & Structure:** 1-minute OHLCV data, millions of rows covering several years
- **Data Types:** Timestamp, Open, High, Low, Close, Volume (all float)
- **Target Variable(s):** None for unsupervised tasks; can predict next-minute/hour price for supervised ML
- **Licensing/Usage Constraints:** Public dataset; academic use allowed

2) Smart Building / IoT Energy Dataset

- **Dataset Name & Source:** [UCI Individual Household Electric Power Consumption](#)
- **Course Topic Alignment:** Week 9: Anomaly Detection, Week 12: Stream Mining
- **Beyond-Course Techniques:**
 - Variational Autoencoder (VAE) for unsupervised anomaly detection
 - Reinforcement Learning for energy optimization
 - LSTM-based forecasting of future consumption
- **Dataset Size & Structure:** ~2M readings, 1-minute frequency over 4 years

- **Data Types:** DateTime, Global_active_power, Global_reactive_power, Voltage, Sub_metering_1–3
- **Target Variable(s):** Optional anomaly labels
- **Licensing/Usage Constraints:** Public; academic use allowed

3) Human Activity Recognition (HAR) Dataset

- **Dataset Name & Source:** [UCI Human Activity Recognition Using Smartphones](#)
- **Course Topic Alignment:** Week 6: Clustering, Week 9: Anomaly Detection
- **Beyond-Course Techniques:**
 - Transformer-based time-series classification
 - Graph-based sensor fusion
 - CNN-based feature learning for sequences
- **Dataset Size & Structure:** 10,299 instances, 561 features, 30 participants
- **Data Types:** Accelerometer/gyroscope features, activity label
- **Target Variable(s):** Activity class (Walking, Sitting, Laying, etc.)
- **Licensing/Usage Constraints:** Public; academic use allowed

(B) Comparative Analysis of Datasets

Dataset	Supported Course Tasks	Data Quality Issues	Algorithmic Feasibility	Bias Considerations	Ethical Considerations
Bitcoin OHLCV	Stream Mining, LSTM/GRU	Missing timestamps (maybe), extreme outliers	High-frequency but manageable; LSTM/Transformer feasible	No participant bias; market microstructure bias possible	No personal data; financial ethics apply
IoT Energy	Anomaly Detection, VAE/LSTM/RL	Missing readings, sensor noise	Medium-size (feasible locally)	Limited household diversity (sample bias)	Household privacy; sensitive energy patterns

Dataset	Supported Course Tasks	Data Quality Issues	Algorithmic Feasibility	Bias Considerations	Ethical Considerations
HAR Sensor	Clustering, CNN/Transformer/Graph	Sensor noise, missing readings	Small dataset (feasible locally)	Limited participants (activity bias)	Minimal; careful with health-related interpretations

(C) Dataset Selection

Dataset: Bitcoin OHLCV Price/Volume Dataset

Reasons:

- Directly supports stream mining, anomaly detection, and large-scale ML (course topics)
- Supports LSTM/Transformer forecasting and change-point detection (beyond-course)
- Real-world, high-frequency, and large-scale enough for a data mining project
- Clean numeric structure so manageable preprocessing and visualization

Trade-offs:

- No graph structure, so cannot explore graph mining or network embeddings
- Limited labeling, so can only do unsupervised or self-supervised tasks

(D) Exploratory Data Analysis (EDA)

Python Notebook:

<https://colab.research.google.com/drive/1mCJXpr5YFjWDJpYa3iUmoewNfwgo8Hol?usp=sharing>

Exploratory Data Analysis (EDA) Results

1. Dataset Overview

- **Number of entries:** 7,422,880 (1-minute intervals)
- **Columns:** Timestamp, Open, High, Low, Close, Volume

- **Memory usage:** 340 MB (can fit in memory comfortably)

2. Numeric Summary

Column	Min	25%	Median	75%	Max	Mean	Std
Open	3.8	447.98	7475.0	34061.44	126202	22182	30721
High	3.8	448.01	7479.93	34079.24	126272	22190	30730
Low	3.8	447.83	7470.01	34040.91	126158	22174	30713
Close	3.8	447.99	7475.0	34061.78	126202	22182	30722
Volume	0.0	0.0203	0.4506	2.8759	5853.85	5.087	21.922

Observations from summary statistics:

- Prices range from **\$3.8 to \$126,202**, showing massive growth and extreme scale differences over time.
- Volumes are highly skewed: most 1-minute windows are very small (< 1 BTC), but occasional spikes reach thousands.
- No missing values - dataset is clean.

3. Time-Series Insights

- **Price trends:** Clear long-term upward trend with spikes and dips (bull/bear cycles).
- **Volatility:** Extreme spikes in log returns and volume correspond to sudden market events.
- **Volume spikes:** Typically coincide with sharp price changes — useful for anomaly detection.

4. Derived Features

- **Log Returns:** $\log_return = \text{np.log}(\text{Close} / \text{Open})$ - Distribution shows most 1-minute returns are tiny, occasional extreme returns indicate anomalies.
- **Rolling Volatility:** Computed over 60-minute windows - Reveals clusters of high market activity (e.g., major price corrections or surges).

(E) Initial Insights and Direction

Observations

1. Most price changes are small in a 1-minute window - extreme returns are rare but impactful.
2. Volume spikes coincide with price swings - strong candidate for anomaly detection.
3. Long-term trends with short-term noise - supports both forecasting and stream-based analysis.

Hypotheses

- High-frequency noise can mask meaningful patterns in short windows → rolling features or smoothing may improve forecasting accuracy.
- Extreme price or volume events may indicate exploitable patterns (anomalies) for downstream analysis.

Potential Research Questions (RQs)

1. How do different rolling-window lengths affect the accuracy of LSTM/Transformer price forecasts?
2. Can anomaly detection on log returns or volume identify periods of market instability?
3. Do technical indicators (RSI, MACD) improve prediction over raw OHLCV features?
4. How do extreme price movements correlate with volume spikes over varying time scales?

(F) GitHub Portfolio Building

- **Repository:** <https://github.com/ShivaniSuresh1/bitcoin-ohlcv-mining>
- **Includes:**
 - EDA notebook with plots, log returns, volatility analysis
 - README describing dataset, EDA and project goals

(G) Collaboration Declaration

On my honor, I declare the following resources:

1. Collaborators:

- None

2. Web Sources:

- <https://pandas.pydata.org/docs/>

- <https://numpy.org/doc/>

- <https://matplotlib.org/stable/>

- <https://www.kaggle.com/datasets/mczielinski/bitcoin-historical-data>

3. AI Tools:

- ChatGPT – Used for structural refinement, documentation formatting and domain specific information.

4. Citations (Papers Referenced Conceptually):

- Mandelbrot, B. (1963). The Variation of Certain Speculative Prices. Journal of Business.: Justifies the observation that log returns exhibit heavy tails and deviate from Gaussian assumptions due to frequent extreme price movements.

- Engle, R. (1982). Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of UK Inflation. Econometrica.: Supports the identification of volatility clustering and time-varying variance in financial return series.

- Bollerslev, T. (1986). Generalized Autoregressive Conditional Heteroskedasticity. Journal of Econometrics.: Motivates the use of rolling volatility and GARCH-type models to capture persistent conditional heteroskedasticity in returns.